# Frequency–Spatial Joint Decoupling With Adaptive Perceptual Aggregation for Remote Sensing Small Object Detection

Ying Gao, Zongshuai Zhang, Zhengyu Zhu, Yuxuan Pang, Junyu Dong, and Yongqing Li, *Member, IEEE*

***Abstract*—Small object detection in remote sensing (RS) imagery remains fundamentally challenging due to severe information degradation caused by limited spatial resolution and complex background interference. In deep neural networks, such degradation is further exacerbated by irreversible information loss during conventional downsampling, leading to weak and ambiguous feature representations. To address this issue, we propose a frequency–spatial joint decoupling and adaptive perceptual aggregation module (FSD-APAM), which explicitly separates signal-level detail preservation and perceptual-level feature discrimination within a unified framework. Specifically, a frequency-domain detail decoupling (FDD) unit leverages discrete wavelet transform to construct reversible feature pathways, enabling the recovery of high-frequency edge information suppressed during downsampling. Complementarily, a spatial salience decoupling (SSD) unit introduces a lateral inhibition mechanism to enhance isolated target responses while suppressing structured background interference. To further ensure global contextual consistency with low computational overhead, an adaptive contextual perceptual aggregation (ACPA) unit is designed to facilitate efficient interaction between sparse target cues and dense semantic representations. Extensive experiments on AI-TODV2, LEVIR-Ship, and VisDrone demonstrate that the proposed method consistently improves detection performance across multiple mainstream architectures without requiring substantial architectural modifications. In particular, FSD-APAM achieves superior accuracy in detecting extremely small objects while maintaining competitive efficiency, highlighting its practical value for large-scale RS applications. The source code is available at https://github.com/cskkx1/FSD-APAM**

***Index Terms*—Decoupled representation learning, lateral inhibition mechanism, remote sensing (RS) images, small object detection, wavelet transform.**

## I. Introduction

With the rapid advancement of Earth observation technologies, high-resolution remote sensing (RS) imagery and uncrewed aerial vehicle (UAV) data have become essential for applications such as urban monitoring, disaster assessment, and maritime surveillance. In these scenarios, accurate detection of small objects, such as vehicles, ships, and pedestrians, is critical for comprehensive scene understanding. Despite the remarkable progress of deep learning-based detectors, such as YOLO [1], Faster R-CNN [2], and RT-DETR [3], their performance degrades significantly when applied to small objects in RS imagery. This limitation primarily arises from the extremely small spatial footprint of targets, which typically occupy only a few pixels and exhibit weak contrast against complex backgrounds.

From a signal processing perspective, small objects in RS imagery suffer from severe information degradation. As shown in datasets such as AI-TODV2 [4] and VisDrone [5], objects smaller than $16 \times 16$ pixels often lack discriminative texture and are easily overwhelmed by noise, atmospheric scattering, and background clutter. More critically, standard downsampling operations in convolutional neural networks, such as pooling or strided convolution, inherently act as low-pass filters. According to the Nyquist–Shannon sampling theorem [6], such operations inevitably suppress high-frequency components, leading to irreversible loss of edge and structural information. Consequently, the already weak features of small objects are further attenuated or even annihilated before reaching high-level semantic layers, limiting the effectiveness of subsequent feature fusion mechanisms.

In addition to signal degradation, the perceptual ambiguity between objects and backgrounds further exacerbates the detection difficulty. RS scenes typically contain large homogeneous regions, such as vegetation, water surfaces, and urban textures, with strong spatial correlations, which can easily produce object-like responses. Existing attention mechanisms attempt to enhance salient features via spatial or channel-wise weighting. However, in scenarios where true targets are extremely sparse, these methods often amplify background interference rather than suppress it. Recent studies such as CIRSM-Net [7] have successfully demonstrated that embedding physical imaging models into deep networks can effectively mitigate signal-level degradation in cross-modal RS tasks. These successes further motivate the integration

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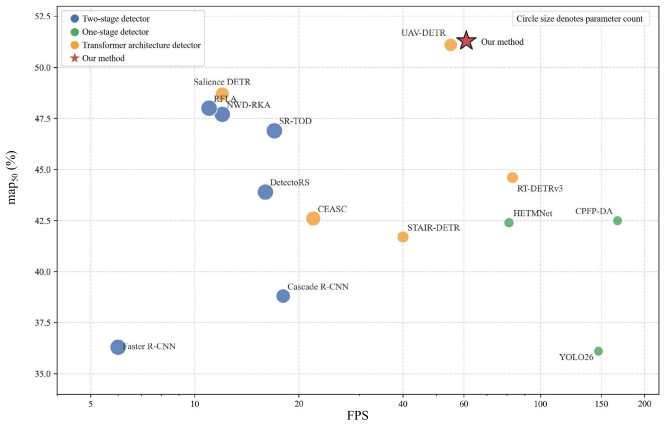



Fig. 1. Speed–accuracy tradeoff comparison on the VisDrone dataset. The bubble size represents the scale of model parameters, and the horizontal axis employs a logarithmic scale for inference speed.

of principled signal processing into single-modal detection pipelines. Nevertheless, the perceptual-level feature entanglement between sparse targets and complex backgrounds in small object detection remains an equally critical challenge that requires explicit disentanglement within a unified framework. This suggests that current approaches lack an explicit mechanism to disentangle object-related features from background-induced responses. These challenges reveal a fundamental limitation of existing methods. They do not explicitly address the dual issues of signal-level information loss and perceptual-level feature entanglement. Inspired by the theory of decoupled representation learning, we argue that improving small object detection requires a unified framework that simultaneously preserves high-frequency signal details and enhances the discriminability of object features from complex backgrounds.

To address these issues, we propose a frequency–spatial joint decoupling and adaptive perceptual aggregation module (FSD-APAM). The proposed module follows a principled design that separates feature processing into complementary stages of recovery, discrimination, and aggregation. Specifically, a frequency-domain detail decoupling (FDD) unit reconstructs high-frequency information via reversible wavelet-based transformations. A spatial salience decoupling (SSD) unit enhances isolated target responses through a biologically inspired lateral inhibition mechanism, effectively suppressing structured background interference. Furthermore, an adaptive contextual perceptual aggregation (ACPA) unit establishes efficient global interactions between sparse target cues and dense semantic representations with linear complexity, ensuring contextual consistency without incurring excessive computational cost. Unlike conventional attention mechanisms that densely model pairwise interactions, ACPA adopts a sparse-to-dense interaction strategy, which significantly reduces computational overhead while preserving global contextual consistency.

To intuitively demonstrate the effectiveness of the proposed method, Fig. 1 presents the speed–accuracy tradeoff on the VisDrone dataset. Rather than claiming absolute lightweight characteristics, we emphasize that the proposed approach achieves a superior balance between detection precision and actual inference speed. Specifically, through its plug-and-play design, it significantly boosts the detection accuracy of efficient one-stage detectors while preserving highly competitive real-time inference speeds, thereby demonstrating strong potential for practical deployment.

The main contributions are summarized as follows.

1) We propose a unified frequency–spatial decoupling framework for small object detection in RS imagery, which explicitly addresses both signal-level information degradation and perceptual-level feature entanglement, providing a principled formulation of the problem.
2) We design three complementary modules, namely FDD, SSD, and ACPA, to perform high-frequency detail recovery, spatial salience discrimination, and efficient global contextual aggregation, respectively, forming a coherent and interpretable processing pipeline.
3) We demonstrate through extensive experiments that the proposed framework can be seamlessly integrated into various detection architectures and consistently improves performance on multiple RS benchmarks, particularly for extremely small object scenarios.

The remainder of this article is organized as follows. Section II provides an overview of related work. Section III introduces the proposed FSD-APAM in detail, highlighting its core architecture and functional units. Section IV presents the experimental evaluation conducted on the AI-TODV2 [4], VisDrone [5], and LEVIR-Ship [8] datasets. Finally, Section V concludes this article based on our findings and discusses potential directions for future research.

## II. Related Work

### A. Small Object Detection Based on Deep Learning

RS object detection has transitioned from handcrafted features to deep representations. Early two-stage architectures, such as Faster R-CNN [2] and Cascade R-CNN [9], established high-precision benchmarks via RPN and cascaded regression. To meet real-time requirements, one-stage algorithms, including the SSD [10] and YOLO [1], [11], [12], [13] series, emerged, achieving a superior balance between accuracy and efficiency through CSPNet backbones and decoupled heads.

Specific to RS, researchers have explored multiscale perception [14] and dual attention mechanisms [15] to suppress background noise. However, for extremely small objects with an area less than 16 $\times$ 16 pixels as defined by AI-TODV2 [4], traditional downsampling in these detectors still leads to severe signal annihilation. Recently, Vision Transformers like DETR [16] and RT-DETR [3], [17], [18], [19] have introduced long-range correlation modeling to enhance semantic understanding. Despite these gains, recovering weak high-frequency details erased at the physical signal level remains a critical bottleneck under limited computational constraints.

### B. Feature Recovery Based on Frequency-Domain Learning and Feature Reconstruction

Traditional CNN downsampling acts as a form of low-pass filtering, which, according to the Nyquist–Shannon sampling

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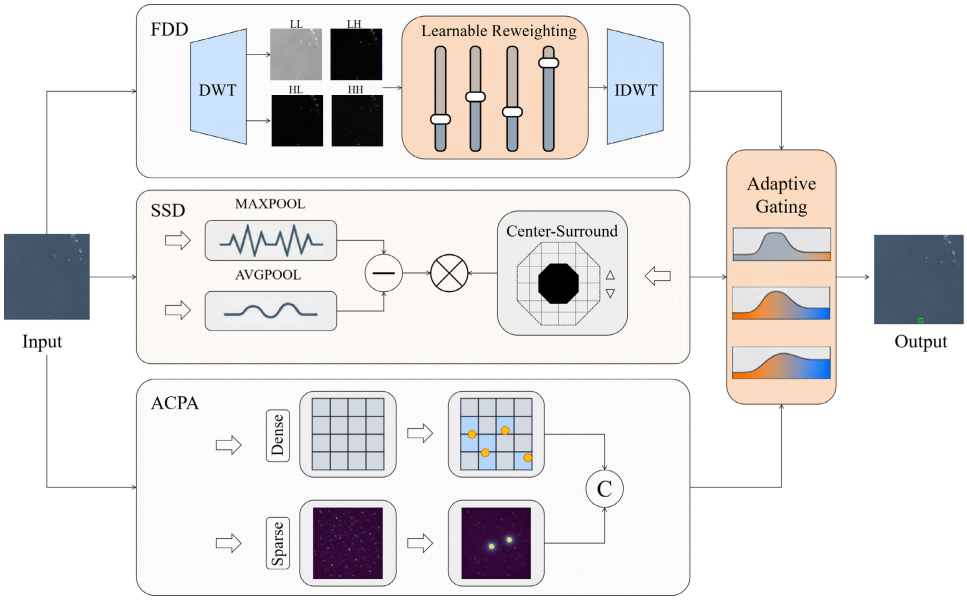



Fig. 2. Overall architectural schematic of FSD-APAM is illustrated below. This module integrates three staged units consisting of FDD, SSD, and ACPA to progressively achieve frequency-domain detail recovery, spatial salience enhancement, and global contextual aggregation.

theorem [6], inevitably leads to feature annihilation for miniature RS objects. Consequently, frequency-domain analysis for refined feature reconstruction has become a critical perspective. As a tool for multiresolution analysis, the discrete wavelet transform [20] decouples signals into approximation and detail components, offering superior capabilities for preserving structure over traditional destructive sampling [20].

In practical applications, WaveCNet [21] demonstrates the efficacy of wavelets in recovering textural details. For detection, frameworks like FFCA-YOLO [22] and ISGLNet [23] enhance weak representations through fusion strategies sensitive to context and intrinsic sensitivity guidance. Inspired by these, embedding wavelet reconstruction into feature extraction networks transforms lost spatial energy into explicit frequency subbands. This mechanism achieves deep compensation from the physical signal level to the semantic level, maximizing the preservation of the target's intrinsic frequency distribution while maintaining efficiency to establish a robust foundation for perception across different scales.

### C. Decoupled Representation Learning and Attention Mechanisms

While deep learning excels at automated feature extraction, traditional CNN features are often highly entangled, making it difficult for models to distinguish targets from complex environmental noise in RS imagery. Decoupled representation learning addresses this by mapping data into a latent space where dimensions control specific factors, utilizing information-theoretic constraints such as the KL divergence in $\beta$-VAE to enhance interpretability and generalization. In RS tasks, feature reconstruction aligns with this decoupling objective to improve discriminative power.

Although general attention mechanisms like CBAM [24], SE-Net [25], and ECA-Net [26] are widely used, they often trigger excessive activation of background clutter in sparse RS scenes with low signal contrast due to global weighting. To mitigate this, researchers have explored reconstruction algorithms incorporating attention; for instance, SCConv [27] reduces redundancy across spatial and channel dimensions, while HFCR-Net [28] and BAFNet [29] employ dual-stream mechanisms and geometric constraints to amplify minute differences. Recent advances in multimodal RS further highlight the value of explicit feature decoupling coupled with adaptive attention. DADIGAN [30] demonstrates this via convolutional sparse coding and progressive dual attention for SAR-optical cloud removal, inspiring us to extend the decoupling-and-aggregation paradigm to single-modal small object detection: the SSD unit performs SSD, while the ACPA unit adaptively aggregates sparse keypoints and dense semantic grids. This article continues this paradigm by introducing biological lateral inhibition and adaptive contextual aggregation to achieve explicit decoupling of targets from background interference, overcoming detection bottlenecks in RS perception.

## III. Methodology

### A. Overview

Small objects in RS face a dual perceptual bottleneck arising from the loss of high-frequency details and complex background interference, as they are highly susceptible to feature annihilation during deep network downsampling [6]. To overcome this, the proposed FSD-APAM architecture (Fig. 2) constructs a parallel decoupling space across multiple views to preserve intrinsic features. Specifically, the FDD unit utilizes DWT [20] to establish a reversible downsampling path, employing learnable spectral reweighting for recovering textures with high fidelity. Simultaneously, the SSD unit uses the difference between max and average pooling, together with lateral inhibition, to suppress nonisolated background noise and identify weak target responses. Complementing these, the ACPA unit facilitates interaction between sparse keypoints and dense semantic grids to balance local energy characteristics with global topological understanding under

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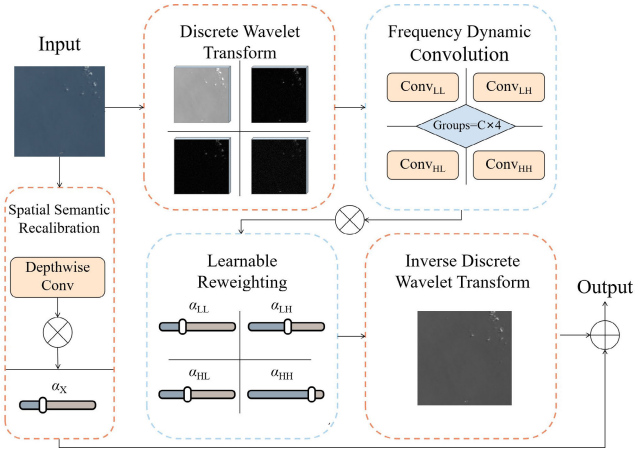



Fig. 3. Schematic of the FDD unit. It comprises spatial semantic recalibration, forward discrete wavelet transform, frequency dynamic convolution, spectral reweighting, and inverse discrete wavelet transform.

$O(HW)$ complexity. Finally, features from these dimensions converge at the adaptive gating fusion unit, where dynamically generated mutually exclusive weights enable the intelligent integration of local details and global semantics, ensuring high sensitivity to tiny targets while effectively suppressing false alarms.

To seamlessly integrate the proposed FSD-APAM into mainstream object detection architectures, we employ a targeted plug-and-play insertion strategy. In CNN-based detectors such as the YOLO series, the FSD-APAM is deployed within the feature fusion network (Neck). Specifically, it is utilized to replace the standard feature processing block at the high-resolution P3 stage following the top-down cross-scale concatenation. Since the P3 scale is primarily responsible for small object prediction, this placement ensures that the fragile high-frequency details and spatial salience of miniature objects are explicitly enhanced at the most critical processing stage. Correspondingly, in Transformer-based architectures like RT-DETR, the module is embedded within the hybrid encoder to refine the spatio-frequency representations of the dense visual tokens before they are processed by the cross-attention decoder.

### B. Frequency-Domain Detail Decoupling

In deep convolutional neural networks, traditional downsampling via strided convolutions or pooling acts as a form of low-pass filtering, which inevitably erases high-frequency components essential for tiny RS objects [6]. Since objects like small vehicles or ships consist almost entirely of signals in the high-frequency domain, this signal smoothing leads to feature annihilation in deep maps. To overcome this, the FDD unit utilizes multiresolution analysis via the DWT [20] to establish a reversible downsampling path. By decomposing features into one low-frequency and three high-frequency detail components through $J$ levels of cascaded filtering, where $J = 1$ by default as detailed in Section IV-D, FDD achieves decoupling in the frequency domain under zero information loss.

As illustrated in Fig. 3, the FDD unit adopts a dual-branch residual structure consisting of a spatial backbone and a frequency-domain branch. The spatial branch calibrates spatial semantic information via $5 \times 5$ depthwise separable convolutions, while the frequency branch handles textural decoupling and reconstruction. First, the input feature map $X \in \mathbb{R}^{C\times H\times W}$ is mapped into frequency space using a 2-D DWT. Given that RS objects often appear as abrupt gray-scale step signals, we select the Haar wavelet [31] as the basis function. The transformation is expressed as

$$\mathbf{X}_{\text{freq}} = \text{DWT}(X) = \begin{bmatrix} (X * f_{LL}) \downarrow_2 \\ (X * f_{LH}) \downarrow_2 \\ (X * f_{HL}) \downarrow_2 \\ (X * f_{HH}) \downarrow_2 \end{bmatrix} \tag{1}$$

where $\mathbf{X}_{\text{freq}}$ stacks four subbands: the approximation $X_{LL}$, and details $X_{LH}, X_{HL}$, and $X_{HH}$ representing horizontal, vertical, and diagonal directions, respectively. Compared with smoother families such as Daubechies [32] or Symlet [33], the short support of the Haar wavelet is highly sensitive to pixel-level discontinuities, naturally mitigating the Gibbs phenomenon [34] and ringing effects. To empirically validate this selection, we evaluated the Haar wavelet against db2 and sym2 wavelets across three diverse RS datasets. Experimental results demonstrate that the Haar basis avoids oversmoothing the weak features of small objects, yielding $mAP_{50}$ improvements of approximately 0.4%, 0.3%, and 0.1% on AI-TODV2, LEVIR-Ship, and VisDrone, respectively, compared to the smoother alternatives. This empirical evidence confirms its suitability for preserving essential edge localization.

To overcome the fixed contribution of subbands in traditional transforms, we introduce a learnable spectral reweighting mechanism. Instead of a uniform scalar, we assign an independent, channel-wise parameter vector $\alpha$ to each distinct high-frequency subband. This allows the network to adaptively calibrate the optimal gain for specific directional textures (horizontal, vertical, and diagonal) while preserving the low-frequency structural base

$$\begin{cases} \tilde{X}_{LL} = X_{LL} \\ \tilde{X}_{LH} = \alpha_{LH} \cdot \text{Conv}_{1\times1}(X_{LH}) \\ \tilde{X}_{HL} = \alpha_{HL} \cdot \text{Conv}_{1\times1}(X_{HL}) \\ \tilde{X}_{HH} = \alpha_{HH} \cdot \text{Conv}_{1\times1}(X_{HH}) \end{cases} \tag{2}$$

$$X_{detail} = \text{IDWT}\left(\tilde{X}_{LL}, \tilde{X}_{LH}, \tilde{X}_{HL}, \tilde{X}_{HH}\right). \tag{3}$$

As analyzed in Section IV-D, assigning a large initial weight introduces severe background noise interference and leads to drastic oscillations during early training stages. Therefore, to balance the recovery of high-frequency details and training stability, we initialize all high-frequency parameter vectors ($\alpha_{LH}, \alpha_{HL}$, and $\alpha_{HH}$) with a constant value of 0.1 across all channels. The enhanced details $X_{detail}$ are then fused with the spatial backbone from the DWConv layer via point-wise addition. This architecture broadens the receptive field while preserving fine textures. The procedure is detailed in Algorithm 1.

By transforming destructive downsampling into a reversible frequency-domain transformation, the FDD unit effectively balances high-level semantics with low-level ground details,

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**Algorithm 1** Algorithm Process for FDD

**Input:** Input feature map $X \in \mathbb{R}^{C\times H\times W}$, learnable scaling factors $\alpha_{LH}, \alpha_{HL}, \alpha_{HH}$
**Output:** Detail-enhanced feature map $Y$

**Step 1: Frequency-domain Decomposition**
$X_{LL}, X_{LH}, X_{HL}, X_{HH} \leftarrow \mathrm{DWT}(X, \text{wavelet} = '\mathit{haar}')$
**Step 2: Spectral Reweighting**
$X_{LH} \leftarrow \mathrm{Conv}(X_{LH}) \cdot \alpha_{LH}$
$X_{HL} \leftarrow \mathrm{Conv}(X_{HL}) \cdot \alpha_{HL}$
$X_{HH} \leftarrow \mathrm{Conv}(X_{HH}) \cdot \alpha_{HH}$
$X_{LL} \leftarrow X_{LL}$
**Step 3: Feature Reconstruction**
$X_{recon} \leftarrow \mathrm{IDWT}(X_{LL}, X_{LH}, X_{HL}, X_{HH})$
**Step 4: Residual Injection**
$Y \leftarrow \mathrm{BaseConv}(X) + X_{recon}$
**return** $Y$

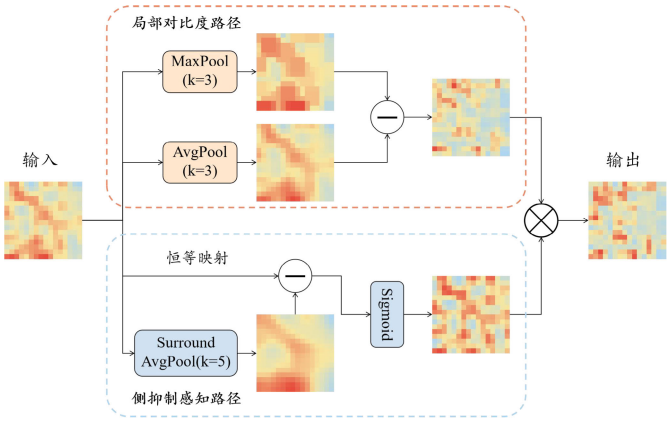



Fig. 4. Schematic of the SSD unit. It comprises a local contrast path and a lateral inhibition perception path.

establishing a robust representation foundation for extremely small RS objects.

### C. Spatial Salience Decoupling

Although the FDD unit effectively restores fine-grained edge and textural details, background regions often dominate complex RS scenes, easily drowning the weak responses of miniature objects. To accurately lock onto small targets while eliminating vast structural clutter, we propose the SSD unit. The architectural design of the SSD unit is deeply inspired by the biological lateral inhibition [35] mechanism inherent in mammalian retinal ganglion cells.

In biological visual networks, a visual neuron responds with high excitation when a stimulus is restricted to the center of its receptive field. However, when the stimulus expands to the surrounding zone, the surrounding pathways laterally inhibit the central excitation, thereby making the neuron highly selective to isolated singularities and insensitive to massive homogeneous textures.

To mathematically formalize this biological behavior within deep feature maps, as illustrated in Fig. 4, the SSD unit constructs a dual-stream architecture comprising a local contrast path and a lateral inhibition path. Specifically, we define a salience mask capable of quantifying the isolation and target probability at each pixel location. For the local contrast path, instead of utilizing global average pooling (GAP), which smooths out weak features, we extract the structural mutations by calculating the local peak-to-average deviation within a neighborhood to generate the contrast feature map $M_{contrast}$

$$M_{contrast}(X) = \mathrm{MaxPool}_{k\times k}(X) - \mathrm{AvgPool}_{k\times k}(X). \quad (4)$$

Simultaneously, to distinguish genuine isolated targets from continuous landforms or structured linear edges, we explicitly formulate the lateral inhibition perception path. We construct an annular receptive field that performs a differential calculation to estimate the local background baseline $M_{isolation}$

$$M_{isolation}(X) = X - DWConv_{k=3}^{(d=2)}(X, W_{annular}) \quad (5)$$

where $DWConv_{k=3}^{(d=2)}(\cdot)$ represents a depthwise dilated convolution expanding the $3 \times 3$ kernel to a physical spatial span of $5 \times 5$ (equivalent to $k+2$). In our specific implementation, to construct the strict annular spatial sampling, we freeze the convolution weights $W_{annular}$, explicitly setting the central weight of this grid to zero and the eight peripheral weights to $1/8$. This pure surrounding sampling operation effectively prevents central target feature leakage into the background estimation. Consequently, isolated tiny targets trigger negligible surround inhibition and achieve a high isolation confidence, whereas continuous structured backgrounds trigger strong lateral inhibition and are effectively annihilated.

**Algorithm 2** Algorithm Process for SSD

**Input:** Feature map $X \in \mathbb{R}^{C\times H\times W}$, kernel size $k$
**Output:** Salience-enhanced feature map $M_s$

**Step 1: Local Contrast Path**
$p \leftarrow \lfloor k/2 \rfloor$
$X_{max} \leftarrow \mathrm{MaxPool2d}(X, \text{kernel} = k, \text{stride} = 1, \text{pad} = p)$
$X_{avg} \leftarrow \mathrm{AvgPool2d}(X, \text{kernel} = k, \text{stride} = 1, \text{pad} = p)$
$M_{contrast} \leftarrow X_{max} - X_{avg}$
**Step 2: Lateral Inhibition Path**
$d \leftarrow 2, \quad p_{sur} \leftarrow d$
$X_{sur} \leftarrow \mathrm{DWConv2d}(X, W_{annular}, \text{kernel} = k, \text{stride} = 1, \text{pad} = p_{sur}, \text{dilation} = d)$
$M_{iso} \leftarrow X - X_{sur}$
**Step 3: Dual-gating Fusion**
$W_{gate} \leftarrow \mathrm{Sigmoid}(M_{iso})$
$M_s \leftarrow M_{contrast} \odot W_{gate}$
**return** $M_s$

Finally, to obtain a high-confidence salience feature map, we conduct a dual-gating fusion of these paths. The Sigmoid function maps the isolation feature $M_{isolation}$ into gating weights to modulate $M_{contrast}$, generating the final spatial salience features $M_s$

$$M_s = M_{contrast} \odot \sigma(M_{isolation}). \quad (6)$$

This output $M_s$ serves as the final decoupled representation of the SSD unit, providing pure spatial guidance. By mathematically simulating lateral inhibition, this design effectively suppresses structured background interference while explicitly locking onto the weak singularities of minute targets. The procedure is detailed in Algorithm 2.

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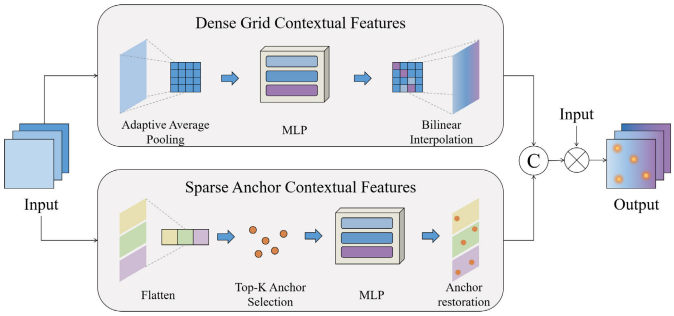



Fig. 5. Schematic of the ACPA unit. It comprises dense grid contextual features and sparse anchor contextual features.

The SSD unit addresses complex backgrounds through explicit spatial mask generation and dual-path processing. Operating alongside the FDD unit, it suppresses structured textural interference while preserving the isolated features of small objects, effectively separating target signals from background noise for subsequent aggregation.

### D. Adaptive Contextual Perceptual Aggregation

To enhance semantic discrimination in complex RS scenarios, incorporating global contextual information is essential, as the identity of an isolated high response point often depends on its surrounding environment. However, existing methods based on self-attention, such as Nonlocal [36] or Transformer [37], exhibit an $O(H^2W^2)$ complexity, which is unaffordable for high-resolution imagery. Conversely, GAP tends to annihilate high-frequency responses of tiny features. To balance efficiency and awareness, we propose the ACPA unit, which constructs a sparse-dense dual-stream interaction mechanism, as shown in Fig. 5.

The ACPA unit abandons uniform global processing for two parallel pathways. The dense semantic stream captures structural information via a gridding strategy. To extract macroscopic geographic topology while mitigating the dimensional disaster associated with dense interaction, the input features $X \in \mathbb{R}^{C\times H\times W}$ are first pooled into a compact $G \times G$ grid for adaptive average pooling. Subsequently, the features undergo semantic interaction via a multilayer perceptron (MLP). To address the potential computational bottleneck caused by the large number of channels in deep features, this MLP is purposefully designed with a bottleneck architecture. It applies a channel reduction ratio of four to compress the features before nonlinear interaction, thereby strictly capping the interaction complexity. Finally, the lightweight interacted features are restored to the original resolution via bilinear interpolation to obtain $X_{dense}$ as follows:

$$X_{dense} = \mathrm{Up}\left(\mathrm{MLP}\left(\mathrm{AdaptiveAvgPool}_{G\times G}\left(X\right)\right)\right). \tag{7}$$

Unlike standard self-attention mechanisms with quadratic complexity, our bilinear upsampling operates with a strictly linear complexity, balancing comprehensive contextual awareness with highly efficient and real-time lightweight performance. In parallel, the sparse keypoint stream utilizes an energy-driven sampling mechanism. We identify the top $K$ significant keypoints $\Omega_{top}$ ($K = \lfloor \gamma \cdot H \cdot W \rfloor$) based on the average spatial energy $E(i,j) = 1/C \sum_{c=1}^{C} |X_{c,i,j}|$. This step acts as a hard-attention filter, where an MLP performs nonlinear enhancement for key positions, while background positions are set to zero

$$X_{sparse}^{(p)} = \begin{cases} \mathrm{MLP}\left(X^{(p)}\right), & \text{if } p \in \Omega_{top} \\ 0, & \text{otherwise.} \end{cases} \tag{8}$$

This sparse mapping strengthens potential object representations while significantly reducing redundancy. To leverage both streams, $X_{dense}$ and $X_{sparse}$ are concatenated to generate contextual gating weights $M_{ctx}$ for adaptive modulation of the original features $X$

$$Y_{ACPA} = X \odot \sigma\left(\mathrm{Conv}_{1\times 1}\left(\left[X_{dense}; X_{sparse}\right]\right)\right). \tag{9}$$

**Algorithm 3** Algorithm Process for ACPA

**Input:** Feature map $X \in \mathbb{R}^{C\times H\times W}$, sparse ratio $\gamma$
**Output:** Context-enhanced feature map $Y_{ACPA}$

**Step 1: Dense Semantic Stream**
$X_g \leftarrow \mathrm{AdaptiveAvgPool2d}(X, \mathrm{out} = 4 \times 4)$
$X_f \leftarrow \mathrm{MLP}(\mathrm{Flatten}(X_g))$
$X_{dense} \leftarrow \mathrm{Upsample}(X_f, \mathrm{size} = H \times W)$
**Step 2: Sparse Keypoint Stream**
$E \leftarrow \mathrm{Mean}(X, \mathrm{dim} = 1)$
$K \leftarrow \lfloor \gamma \cdot H \cdot W \rfloor$
$\mathcal{I}_{top} \leftarrow \mathrm{TopK}(E, K)$
$X_{sparse} \leftarrow \mathrm{Zeros_like}(X)$
$X_{sparse}[\mathcal{I}_{top}] \leftarrow \mathrm{MLP}(X[\mathcal{I}_{top}])$
**Step 3: Dual-stream Fusion and Gating**
$X_{fused} \leftarrow \mathrm{Concat}([X_{dense}, X_{sparse}], \mathrm{dim} = 1)$
$M_{ctx} \leftarrow \mathrm{Sigmoid}(\mathrm{Conv}_{1\times 1}(X_{fused}))$
$Y_{ACPA} \leftarrow X \odot M_{ctx}$
**return** $Y_{ACPA}$

Through this mechanism, the ACPA unit captures global geospatial topological relationships without overwhelming salient small object features with background noise. This process achieves comprehensive perception with $O(HW)$ complexity, as summarized in Algorithm 3.

The ACPA unit synergizes dense structural preservation and sparse energy sampling to provide contextual awareness that balances efficiency and precision, circumventing the bottlenecks of traditional attention mechanisms.

### E. Adaptive Gating Fusion Unit

To integrate frequency-domain details, spatial salience, and global context into a unified representation, we design the adaptive gating fusion unit. Instead of simple addition, this unit employs a dual-stream competition mechanism between local and global information using dynamically generated mutually exclusive weights.

First, base convolutional features $X_{base}$, frequency enhanced features $X_{detail}$, and spatial salience features $X_{saliency}$ are concatenated and integrated via a $1 \times 1$ convolution to construct the local feature stream $L$

$$L = \mathrm{Conv}_{1\times 1}\left(\mathrm{Concat}\left(\left[X_{base}, X_{detail}, X_{saliency}\right]\right)\right). \tag{10}$$

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Simultaneously, this local feature stream is processed by the ACPA unit to generate the global feature stream $G$

$$G = \mathrm{ACPA}(L). \tag{11}$$

To prevent local weak signals from being overwhelmed by background noise, a dynamic gating mechanism based on channel attention is introduced. GAP compresses the spatial dimensions of $L$ and $G$ into channel vectors, and an MLP predicts mutually exclusive weight vectors $W_L$ and $W_G$, where $W_L + W_G = 1$. The final output $Y_{final}$ is obtained via weighted summation

$$\begin{bmatrix} W_L & W_G \end{bmatrix} = \mathrm{Softmax}\big(\mathrm{MLP}\big(\mathrm{Concat}[\mathrm{GAP}(L), \mathrm{GAP}(G)]\big)\big) \tag{12}$$

$$Y_{final} = W_L \odot L + W_G \odot G. \tag{13}$$

This mechanism allows the network to adaptively shift focus during inference. In complex textures, the model prioritizes $G$ for global semantic validation, while at object edges, it emphasizes local details from $L$. This dynamic adjustment ensures high sensitivity to tiny targets while effectively suppressing false alarms in complex geographic backgrounds.

## IV. Experiment

### A. Experimental Setup

*1) Datasets:* To verify the effectiveness of the proposed method, this article conducts experiments on three datasets, including AI-TODV2 [4], VisDrone2019 [5], and LEVIR-Ship [8].

AI-TODV2 is an optical RS dataset specifically designed for small object detection. It contains a total of 28 036 images with a size of 800 × 800 pixels, which are partitioned into training, validation, and testing sets with 11 214, 2804, and 14 018 images, respectively. AI-TODV2 covers eight common object categories including airplane, bridge, storage tank, ship, swimming pool, vehicle, person, and windmill. The average object size in the AI-TODV2 dataset is 18.2 pixels, which is significantly smaller than benchmarks in natural scenes such as MS COCO at 99.5 pixels and PASCAL VOC at 156.6 pixels. Such minute object sizes pose a tremendous challenge to object detection algorithms when applied to optical RS images.

LEVIR-Ship is a large-scale dataset dedicated to the detection of miniature ships in RS imagery. It comprises 3896 images, each with a resolution of 512 × 512 pixels, and covers over 3000 miniature ship instances. The pixel sizes of the ship objects are mostly smaller than 20 × 20 and are concentrated around 10 × 10 pixels, making it particularly suitable for small object detection tasks in RS scenarios.

VisDrone2019 focuses on scenarios captured by UAVs and contains a total of 8599 images. The dataset is divided into training, validation, and testing sets, which contain 6471, 548, and 1580 images, respectively. It provides annotations for 11 different categories, including pedestrian, person, bicycle, car, van, truck, tricycle, awning-tricycle, bus, and motor. A unique advantage of the VisDrone dataset lies in its diversity regarding locations, environments, objects, and densities.

*2) Implementation Details:* All experiments are conducted in a distributed environment featuring two NVIDIA GeForce RTX 4090 GPUs and an Intel Xeon Platinum 8352V CPU. The software setup includes PaddlePaddle 2.6.1 and PyTorch 2.3.0, supported by CUDA 12.1 and the nvcc 12.1.105 compiler. This dual-framework configuration ensures experimental flexibility and cross-platform validation.

To ensure reproducibility, random seeds are fixed at 42. Training employs the SGD optimizer with a momentum of 0.937 and a weight decay of 0.0005. The initial learning rate of 0.01 is adjusted via a cosine annealing strategy, with a batch size of 16. Data augmentation techniques, such as Mosaic, random horizontal flipping, and HSV perturbation, are introduced to prevent overfitting. During inference, NMS is applied with an IoU threshold of 0.65 and a confidence threshold of 0.001.

*3) Evaluation Metrics:* To quantitatively evaluate the model, we adopt the MS COCO standard evaluation system. Primary metrics include $mAP_{50}$ and $mAP_{50:95}$, which reflect detection robustness across various IoU thresholds. Specifically, we utilize the $AP_s$ metric for objects with an area less than $32^2$ pixels to verify the effectiveness in capturing minute object details. Furthermore, the number of parameters and GFLOPs are employed to evaluate spatial and computational complexity, while inference speed is measured in FPS to assess the model's real-time operational efficiency on actual hardware.

### B. Comparison With SOTA Methods

To comprehensively verify the effectiveness and advancement of FSD-APAM, we conduct a comparative evaluation against various object detection models on the AI-TODV2, LEVIR-Ship, and VisDrone datasets. The baseline includes several SOTA RS algorithms specifically optimized for small object detection.

*1) Experiment on AI-TODV2:* As shown in Table I, our method achieves an $mAP_{50}$ of 61.2% on the AI-TODV2 dataset, significantly outperforming RS-specific algorithms such as BAFNet [29]. To evaluate the model's performance across scales, we utilize multiscale metrics including $AP_{vt}$ for objects ranging from $2^2$ to $8^2$ pixels, $AP_t$ for objects ranging from $8^2$ to $16^2$ pixels, $AP_s$ for objects ranging from $16^2$ to $32^2$ pixels, and $AP_m$ for objects ranging from $32^2$ to $64^2$ pixels.

Experimental results indicate that our method achieves 16.6% on the $AP_{vt}$ metric, substantially outperforming other models optimized for tiny scales, such as DetectoRS [38] with NWD [39] and RFLA [40], which yield 6.2% and 8.5%, respectively. This performance underscores the advantage of the FDD module in capturing dim targets at the pixel level through frequency-domain restoration. Furthermore, 28.7% achieved in $AP_t$ reflects the synergy between SSD's lateral inhibition and ACPA's global validation in suppressing background false alarms. Across the YOLO series, FSD-APAM achieves consistent $mAP_{50}$ improvements ranging from 10.5% to 10.8%, further demonstrating the model's exceptional robustness in recovering degraded features.

*2) Experiment on LEVIR-Ship:* Experimental results on the LEVIR-Ship dataset, presented in Table II, validate the

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TABLE I
PERFORMANCE COMPARISON OF DIFFERENT DETECTION ARCHITECTURES ON AI-TODV2 DATASET

| Method | $mAP_{50}$ | $mAP_{50-95}$ | $AP_{vt}$ | $AP_t$ | $AP_s$ | $AP_m$ |
|---|---|---|---|---|---|---|
| SSD [10] | 21.0 | 6.5 | 1.0 | 5.3 | 11.9 | 21.2 |
| Faster R-CNN [2] | 22.1 | 9.7 | 0.0 | 5.2 | 21.7 | 33.1 |
| Cascade R-CNN [9] | 28.6 | 12.8 | 0.1 | 8.8 | 25.0 | 35.2 |
| DetectoRS [38] | 32.2 | 14.5 | 0.1 | 10.6 | 27.7 | 37.7 |
| DetectoRS + NWD [39] | 47.6 | 19.3 | 6.2 | 18.9 | 28.6 | 37.5 |
| DetectoRS + RFLA [40] | 53.3 | 17.5 | 8.5 | 23.4 | 29.9 | 37.7 |
| FCOS [41] | 24.2 | 9.8 | 1.2 | 7.8 | 16.1 | 27.2 |
| YOLOv8 [12] | 29.5 | 13.2 | 4.8 | 15.2 | 19.5 | 26.8 |
| **YOLOv8 + ours** | 40.2 | 18.1 | 6.5 | 17.8 | 22.3 | 28.5 |
| YOLOv11 [13] | 30.7 | 14.0 | 5.1 | 16.5 | 20.6 | 27.8 |
| **YOLOv11 + ours** | 41.5 | 18.8 | 6.8 | 18.4 | 22.9 | 29.0 |
| YOLOv26 [1] | 32.0 | 14.6 | 5.5 | 17.0 | 21.2 | 28.5 |
| **YOLOv26 + ours** | 42.5 | 19.1 | 7.2 | 19.1 | 23.5 | 29.8 |
| RT-DETRv1 [17] | 51.0 | 19.2 | 6.5 | 19.0 | 23.0 | 29.0 |
| **RT-DETRv1 + ours** | 55.8 | 23.5 | 14.5 | 25.0 | 27.5 | 35.0 |
| RT-DETRv2 [18] | 54.0 | 20.0 | 7.5 | 20.0 | 24.5 | 30.5 |
| **RT-DETRv2 + ours** | 59.0 | 24.5 | 15.5 | 26.5 | 28.0 | 37.5 |
| RT-DETRv3 [19] | 56.7 | 20.9 | 8.8 | 21.0 | 25.4 | 31.5 |
| **RT-DETRv3 + ours** | **61.2** | 25.3 | **16.6** | **28.7** | 28.4 | **39.1** |
| DNTR [42] | 56.7 | 26.2 | 12.8 | 26.4 | 31.0 | 37.1 |
| HANet [43] | 53.7 | 22.1 | 10.9 | 22.2 | 27.3 | 31.5 |
| BAFNet [29] | 59.5 | **28.5** | 16.4 | 28.6 | **32.1** | 38.5 |
| RFHA-YOLO [44] | 56.3 | 26.5 | 14.1 | 26.8 | 30.3 | 35.1 |

TABLE II
PERFORMANCE COMPARISON OF DIFFERENT DETECTION ARCHITECTURES ON LEVIR-SHIP DATASET

| Method | $mAP_{50}$ | $mAP_{50-95}$ | Method | $mAP_{50}$ | $mAP_{50-95}$ |
|---|---|---|---|---|---|
| SSD [10] | 52.9 | 22.3 | Faster R-CNN [2] | 70.8 | 27.6 |
| Efficient-D0 [45] | 71.3 | 27.9 | RetinaNet [46] | 74.9 | 30.5 |
| ORFENet [47] | 81.3 | 32.2 | Efficient-D2 [45] | 80.9 | 31.8 |
| FCOS [41] | 75.5 | 30.5 | YOLOv8 [12] | 74.0 | 31.2 |
| YOLOv11 [13] | 76.2 | 32.5 | YOLOv26 [1] | 78.8 | 31.5 |
| CenterNet [48] | 77.7 | 30.9 | DRENet [49] | 79.5 | 31.3 |
| RT-DETRv1 [17] | 79.0 | 31.8 | RT-DETRv2 [18] | 80.2 | 32.2 |
| RT-DETRv3 [19] | 81.6 | 32.6 | BAFNet [29] | 82.2 | 33.5 |
| FFCA-YOLO [22] | 81.3 | 32.5 | **YOLOv8 + ours** | 76.5 | 33.5 |
| **YOLOv11 + ours** | 78.5 | 34.5 | **YOLO26 + ours** | 80.6 | 32.8 |
| **RT-DETRv1 + ours** | 81.5 | 33.2 | **RT-DETRv2 + ours** | 82.8 | 33.5 |
| **RT-DETRv3 + ours** | **84.4** | **34.3** | | | |

effectiveness and architectural flexibility of the FSD-APAM module. Across the YOLO series, FSD-APAM achieves an average $mAP_{50}$ improvement of 2.2%, while the RT-DETR series gains an average of 2.7%. Notably, with RT-DETRv3 as the baseline, $mAP_{50}$ increases from 81.6% to 84.4%, demonstrating that the frequency–spatial decoupling mechanism effectively compensates for low-level representation deficiencies in processing RS small objects.

Horizontal comparisons show our method outperforming classical models like Faster R-CNN [2] and RetinaNet [46], as well as recent SOTA RS models like FFCA-YOLO [22] and BAFNet [29]. This confirms that frequency-domain restoration and SSD achieve robust, high-precision detection in maritime monitoring.

Visual comparisons in Fig. 6 highlight the superiority of the proposed scheme in complex maritime scenarios.

Regarding recall, rows 1 and 3 of Fig. 6 show that while comparative algorithms miss dim, miniature ships, our model successfully captures pixel-level targets by restoring high-frequency details via the FDD module. For false alarm suppression, rows 2 and 4, the SSD module's lateral inhibition and ACPA's topological validation effectively eliminate false responses from near-shore clutter and waves. Furthermore, enhanced salience centering leads to superior edge fitting in detection boxes, validating the robustness of our architecture.

*3) Experiment on VisDrone:* To verify compatibility, we integrate FSD-APAM into various host networks, including the YOLO series [1], [12], [13] and DETR series [16], [17], [18], [19]. Comparative tests are conducted on the challenging VisDrone dataset, characterized by high viewpoints and dense small object distributions.

Table III indicates that FSD-APAM consistently boosts performance across different architectures. For lightweight CNNs including YOLOv8n, YOLOv11n, and YOLOv26n as cited in [1], [12], and [13], $mAP_{50}$ increases by up to 4.5%. Notably, the improved YOLOv26n maintains a high inference speed of 118 frames/s. In Transformer-based architectures as described in [17], the module elevates the RT-DETRv3 baseline [19] to 51.3% $mAP_{50}$ with a 4.7% gain. This effectively mitigates feature annihilation while preserving real-time performance at 61 frames/s.

At the same time, Table III compares our method with SOTA algorithms. While traditional two-stage methods like RFLA [40] reach 48.0% $mAP_{50}$ with a large number of

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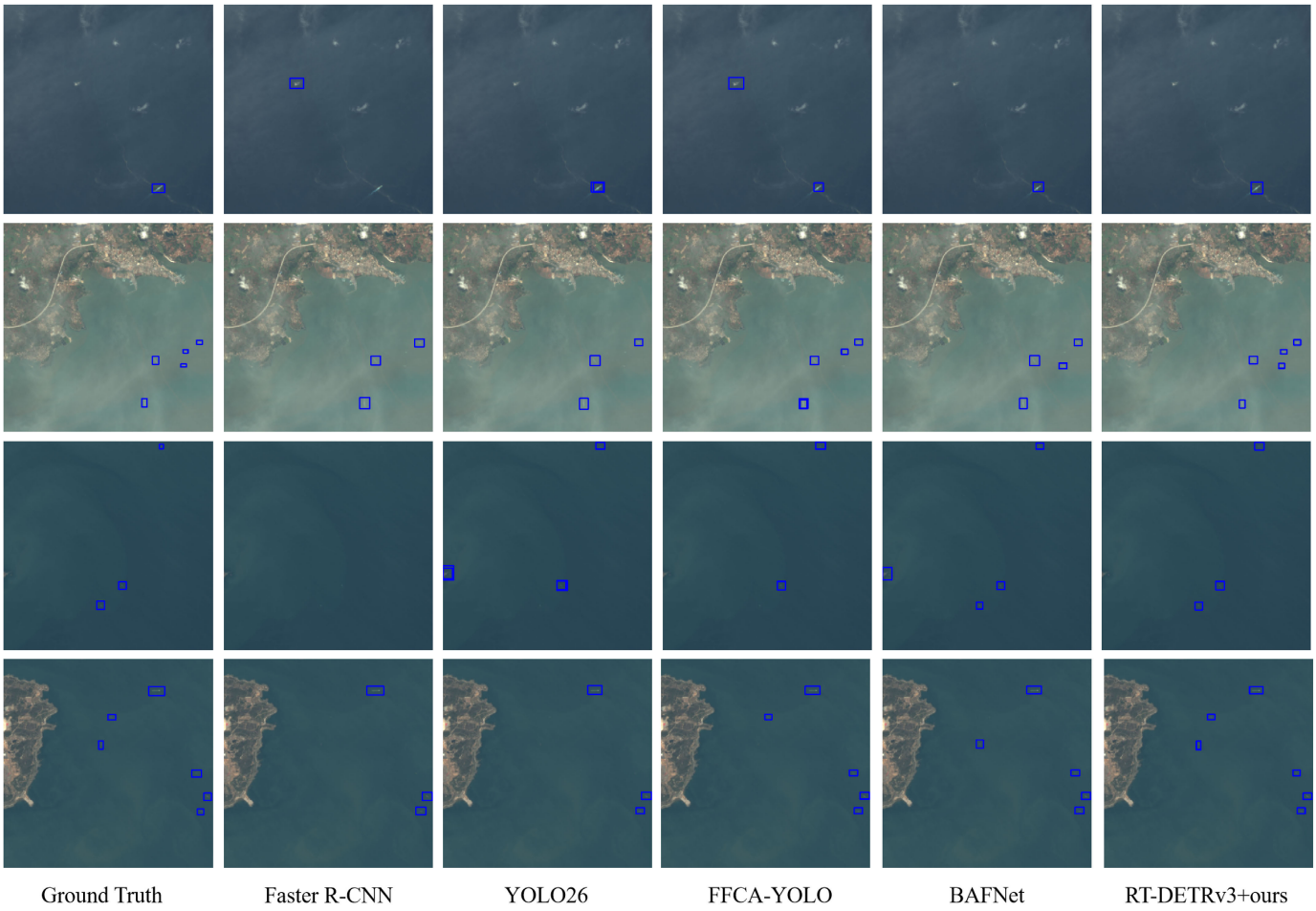



Fig. 6. Visual comparison of detection results on the LEVIR-Ship dataset. The columns correspond to ground truth, cascade R-CNN, YOLO11, DetectoRS + RFLA, BAFNet, and RT-DETRv3 + ours.

parameters, our model achieves 51.3% $mAP_{50}$, surpassing both NWD [39] and the UAV-specific UAV-DETR [54]. In terms of efficiency, though lightweight one-stage models [1], [50] exceed 140 frames/s, they suffer from precision bottlenecks. Conversely, our method balances accuracy and speed, using only 54% of UAV-DETR's parameters and 40% of its computational complexity.

### C. Ablation Study

To systematically verify the independent contributions and synergistic effects of the FDD, SSD, and ACPA units, this article conducts layer-by-layer ablation experiments. With YOLO11n [13] as the unified baseline, these experiments are performed on three datasets exhibiting diverse RS characteristics: AI-TODV2 for extremely small objects, LEVIR-Ship for typical maritime targets, and VisDrone for complex aerial scenarios. The experimental results are summarized in Table IV.

Furthermore, to intuitively reveal the specific impact of each core component, we perform a layer-by-layer visual ablation analysis on the VisDrone dataset using Grad-CAM. Fig. 7 illustrates the evolution of feature attention, reflecting the individual visual contributions and synergistic enhancement of the FDD, SSD, and ACPA modules.

*1) Effectiveness of the FDD:* Table IV indicates that FDD significantly improves the detection of extremely small objects. On AI-TODV2, $mAP_{50}$ rises by 7.5% after introducing FDD, the largest increment among all single module schemes. This demonstrates that restoring high-frequency edge information via DWT is more effective at avoiding feature annihilation than spatial downsampling. As shown in Fig. 7(c), FDD effectively recalls dim and weak targets originally overwhelmed by the background, particularly at frame edges and distant points.

*2) Effectiveness of the SSD:* The SSD module excels in optimizing localization accuracy. As shown in Table IV, $mAP_{50-95}$ reaches 17.6% on AI-TODV2 with SSD alone, superior to FDD's 16.6%. In complex VisDrone scenarios, SSD provides a +1.2% $mAP_{50}$ gain by suppressing nonisolated background interference through the biological lateral inhibition mechanism. Fig. 7(d) confirms that SSD enables thermal response points to fit actual physical contours more closely, enhancing detection robustness for marginal and occluded objects.

*3) Effectiveness of the ACPA:* ACPA demonstrates strong adaptability for dense distributions by capturing global geographic context. It increases VisDrone $mAP_{50}$ to 36.7%, the highest among the three modules. This sparse-dense interaction mechanism eliminates false alarms caused by geographic logic errors. As illustrated in Fig. 7(e), ACPA deepens the activation intensity at target centers while suppressing

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TABLE III
PERFORMANCE COMPARISON OF DIFFERENT DETECTION ARCHITECTURES ON VISDRONE DATASET

| Method | $mAP_{50}$ | $mAP_{50-95}$ | Params (M) | GFLOPs | FPS |
|---|---|---|---|---|---|
| *Two-stage Detectors* | | | | | |
| Faster R-CNN [2] | 36.3 | 21.5 | 137.0 | 271.7 | 6 |
| Cascade R-CNN [9] | 38.8 | 22.6 | 68.9 | 236.4 | 18 |
| DetectoRS [38] | 43.9 | 26.3 | 134.8 | 160.0 | 16 |
| NWD [39] | 47.7 | 27.2 | 135.2 | 278.4 | 12 |
| RFLA [40] | 48.0 | 27.2 | 142.7 | 267.1 | 11 |
| SR-TOD [50] | 46.9 | 27.3 | 138.2 | 251.5 | 17 |
| *One-stage Detectors* | | | | | |
| HETMNet | 42.4 | 23.1 | 4.8 | 20.4 | 81 |
| CPFP-DA [51] | 42.5 | 25.8 | 3.01 | 8.2 | 167 |
| YOLOv8n [12] | 33.2 | 19.3 | 3.01 | 8.2 | 167 |
| YOLOv11n [13] | 34.6 | 19.9 | 2.91 | 7.7 | 155 |
| YOLO26 [1] | 36.1 | 20.8 | 2.51 | 5.8 | 147 |
| *Transformer-based Detectors* | | | | | |
| CEASC [52] | 42.6 | 25.1 | 87.3 | 158.2 | 22 |
| DETR [16] | 34.5 | 19.2 | 40.95 | 86.15 | 28 |
| Salience DETR [53] | 48.7 | 28.4 | 56.5 | 168.0 | 12 |
| STAIR-DETR [54] | 41.7 | 23.4 | 21.2 | 86.6 | 40 |
| UAV-DETR [55] | 51.1 | 28.3 | 42.0 | 170.0 | 55 |
| RT-DETRv1 [17] | 42.0 | 24.6 | 19.03 | 60.23 | 82 |
| RT-DETRv2 [18] | 43.0 | 25.2 | 19.03 | 60.23 | 82 |
| RT-DETRv3 [19] | 46.6 | 28.2 | 19.03 | 60.23 | 83 |
| *Our Enhanced Models* | | | | | |
| YOLOv8n + ours | 37.7 ↑4.5 | 21.3 ↑2.0 | 7.31 | 39.1 | 131 |
| YOLOv11n + ours | 38.3 ↑3.7 | 21.7 ↑1.8 | 7.21 | 38.6 | 120 |
| YOLOv26n + ours | 40.4 ↑4.3 | 22.9 ↑2.1 | 5.30 | 27.1 | 118 |
| DETR + ours | 37.6 ↑3.1 | 20.2 ↑1.0 | 44.30 | 94.7 | 25 |
| RT-DETRv1 + ours | 45.0 ↑3.0 | 25.9 ↑1.3 | 22.68 | 68.7 | 60 |
| RT-DETRv2 + ours | 47.9 ↑4.9 | 26.9 ↑1.7 | 22.68 | 68.7 | 60 |
| **RT-DETRv3 + ours** | **51.3** ↑4.7 | **28.3** ↑0.1 | 22.68 | 68.7 | 61 |

TABLE IV
PERFORMANCE COMPARISON OF DIFFERENT MODULE COMBINATIONS ON AI-TODV2, LEVIR-SHIP, AND VISDRONE DATASETS

| YOLO11 | FDD | SSD | ACPA | AI-TODV2 [4] | | LEVIR-Ship [8] | | VisDrone [5] | |
|---|---|---|---|---|---|---|---|---|---|
| | | | | $mAP_{50}$ | $mAP_{50-95}$ | $mAP_{50}$ | $mAP_{50-95}$ | $mAP_{50}$ | $mAP_{50-95}$ |
| ✓ | – | – | – | 30.7 | 14.0 | 76.2 | 32.5 | 34.6 | 19.9 |
| ✓ | ✓ | – | – | 38.2 | 16.6 | 78.7 | 32.8 | 35.2 | 20.2 |
| ✓ | – | ✓ | – | 37.8 | 17.6 | 77.5 | 32.9 | 35.8 | 20.6 |
| ✓ | – | – | ✓ | 36.8 | 16.9 | 77.1 | 32.6 | 36.7 | 21.1 |
| ✓ | ✓ | ✓ | ✓ | **41.5** | **18.8** | **79.8** | **33.3** | **38.3** | **21.7** |

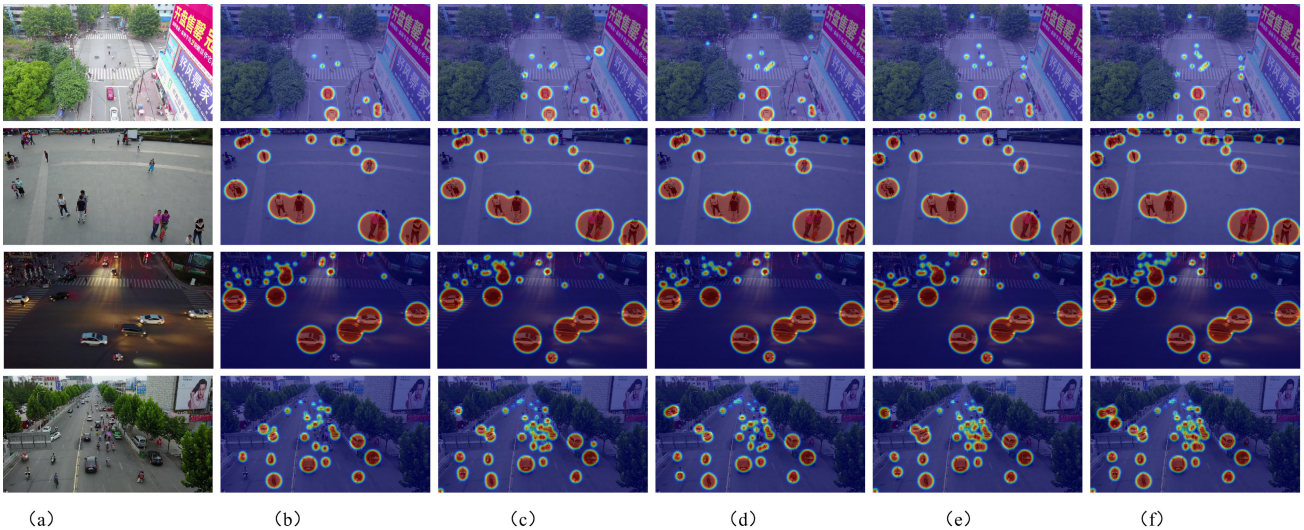



Fig. 7. Ablation visualization of heatmaps with progressive module integration. (a) Input image, (b) baseline YOLO11n, (c) FDD, (d) SSD, (e) ACPA, and (f) all three modules.

surrounding noise, effectively reducing feature adhesion and misjudgment.

*4) Effectiveness of the FSD-APAM:* Collaborative integration of the three modules achieves optimal performance across

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TABLE V
IMPACT OF ACPA SPARSE RATIO $\gamma$ ON PERFORMANCE AND EFFICIENCY

| $\gamma$ | $mAP_{50}$ | $mAP_{50-95}$ | $AP_S$ | GFLOPs | FPS |
|---|---|---|---|---|---|
| 0.01 | 35.2 | 19.8 | 12.5 | **38.1** | **120** |
| 0.03 | 37.1 | 20.9 | 15.3 | 38.3 | 122 |
| **0.05** | **38.3** | **21.7** | **16.8** | 38.6 | **120** |
| 0.10 | 37.8 | 21.3 | 16.1 | 39.5 | 115 |
| 0.20 | 37.1 | 20.8 | 15.4 | 41.2 | 108 |

TABLE VI
IMPACT OF FDD WAVELET DECOMPOSITION LEVELS $J$ ON DETECTION PERFORMANCE

| $J$ | $mAP_{50}$ | $AP_S$ | $AP_M$ | Params (M) |
|---|---|---|---|---|
| **1** | **38.3** | **16.8** | 28.5 | **7.21** |
| 2 | 37.6 | 15.9 | **29.1** | 7.35 |
| 3 | 36.8 | 14.6 | 28.4 | 7.58 |

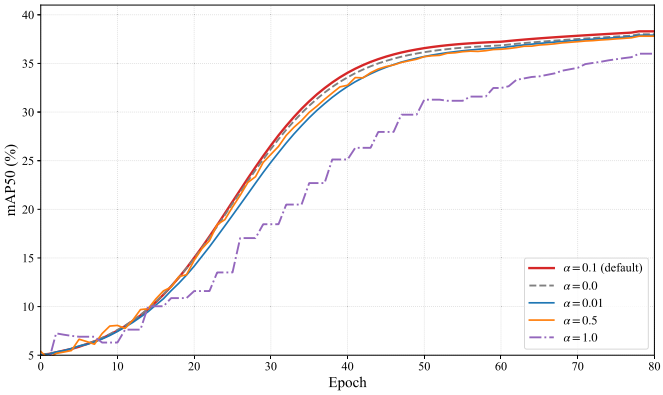



Fig. 8. Comparison of validation accuracy convergence curves under different initial reweighting factors $\alpha$.

all metrics. FDD, SSD, and ACPA focus on signal retrieval, background filtering, and semantic validation, respectively. This parallel decoupling across multiple granularities significantly enhances feature perception for RS small objects. Fig. 7(f) illustrates that the complete model balances anti-interference, high recall, and precise localization, resulting in superior signal-to-noise ratios in the feature maps.

### D. Hyperparameter Sensitivity Analysis

Several key hyperparameters in the FSD-APAM module are analyzed on the VisDrone dataset [5] to balance detection precision and computational efficiency. This experiment focuses on three core variables: the sparse ratio $\gamma$ in ACPA, the wavelet decomposition levels $J$ in FDD, and the initial spectrum reweighting factor $\alpha$.

The sparse ratio $\gamma$ dictates the proportion of high-energy points involved in global attention calculation. As shown in Table V, $AP_S$ climbs from 12.5% to a peak of 16.8% as $\gamma$ increases from 0.01 to 0.05, successfully capturing low-contrast small objects. However, further increasing $\gamma$ to 0.20 introduces contextual noise and reduces inference speed to 108 frames/s. Thus, $\gamma = 0.05$ is selected as the optimal setting.

Regarding the FDD unit, $J$ determines the frequency-domain depth of the receptive field. Results in Table VI demonstrate that the model achieves peak precision for small objects at $J = 1$. As $J$ increases, $AP_S$ exhibits a clear downward trend, dropping to 14.6% at $J = 3$, confirming that high-frequency features are primarily concentrated in shallow wavelet coefficients. Thus, $J = 1$ is the optimal choice to balance feature fidelity and efficiency.

Finally, the impact of the initial spectrum reweighting factor $\alpha$ on model convergence is explored. As illustrated in Fig. 8, $\alpha = 0.1$ ensures training stability and yields the highest precision. Conversely, setting $\alpha = 1.0$ leads to severe oscillations in the early stages due to background noise interference, resulting in lower final precision than the baseline without using high-frequency features, where $\alpha = 0.0$.

## V. CONCLUSION

This article presents FSD-APAM, a unified frequency–spatial decoupling framework for small object detection in RS imagery. By explicitly addressing signal-level information degradation and perceptual-level feature entanglement, the proposed method effectively enhances the representation of extremely small objects under complex background conditions. The FDD, SSD, and ACPA units collaboratively enable high-frequency detail recovery, spatial salience discrimination, and efficient contextual aggregation within a coherent pipeline.

Extensive experiments on multiple benchmarks demonstrate that the proposed method consistently improves detection performance across different architectures, particularly for extremely small objects, while maintaining competitive efficiency. Future work will explore extending the proposed framework to multimodal and spatio-temporal scenarios, as well as developing more adaptive mechanisms for dynamic feature decoupling in real-world RS applications.

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